

Adaptive Kernel Selection for Stein Variational Gradient Descent

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Abstract

Efficient approximation of intractable posterior distributions is a central problem in Bayesian inference. Stein Variational Gradient Descent (SVGD) addresses it as a deterministic variational method that transports an ensemble of particles toward a target distribution. Because the SVGD dynamics are governed by a reproducing kernel Hilbert space (RKHS), their behavior is decisively shaped by the kernel function, which controls both the rate of convergence and the fidelity of the resulting approximation. The median heuristic, though widely adopted for setting the kernel bandwidth, is a single-scalar rule that lacks the flexibility to adapt to the target and degrades markedly as the dimension grows. We depart from this fixed-kernel practice and instead select kernel parameters adaptively over an arbitrary parameterized family of kernels. Recent convergence analyses framed in terms of the kernelized Stein discrepancy (KSD) indicate that maximizing the KSD over the kernel is precisely the criterion that accelerates the underlying dynamics. Building on this principle, we introduce Adaptive SVGD (Ad-SVG), which interleaves SVGD particle transport with gradient ascent on the KSD with respect to the kernel parameters, thereby tuning the kernel online at negligible additional cost. We establish a theoretical analysis that lifts existing convergence guarantees for KSD minimization under a fixed kernel to the adaptive regime, proving that the maximal KSD over the kernel class vanishes along the iterates and, under a Stein log-Sobolev inequality, quantifying convergence in KL divergence together with explicit iteration-complexity and bias bounds. These guarantees are borne out empirically: across Gaussian mixtures, an ODE-governed linear inverse problem, high-dimensional Gaussian and Gaussian-process targets, and Bayesian logistic regression, Ad-SVG consistently surpasses the median heuristic and, in particular, alleviates the variance collapse that afflicts SVGD as dimension increases.

1 Introduction

Stein Variational Gradient Descent (SVGD) [SVGD] is a deterministic, particle-based method for approximate Bayesian inference that has established itself as a compelling alternative to Markov Chain Monte Carlo (MCMC). SVGD evolves an ensemble of particles along update directions given by the functional gradient of the Kullback–Leibler (KL) divergence to the target, with the admissible directions constrained to the unit ball of a reproducing kernel Hilbert space (RKHS). This construction couples the method’s efficacy tightly to the geometry of that space, and hence to the kernel: kernel choice governs the interaction between particles and, through it, both the rate of convergence and the quality of the approximation [Duncan2019OnTG, Nsken2021SteinVG]. A well-documented symptom of a poorly chosen kernel is that the particle approximation systematically underestimates the posterior variance [DBLP:conf/iclr/BaEGSSWZ22]. Observations of this kind have hardened into a widely held belief that SVGD is inherently ill-suited to high-dimensional targets. We contend that this apparent limitation is largely an artifact of a fixed,

hand-tuned kernel rather than of the dynamics themselves. To substantiate this claim, we introduce an adaptive mechanism that tunes the kernel parameters online during inference by maximizing the kernelized Stein discrepancy (KSD), thereby restoring effective particle transport in complex and high-dimensional spaces.

1.1 Related work.

Since its introduction [SVGD], SVGD has become a widely used tool for approximate Bayesian inference across a broad range of machine learning applications [DBLP:conf/uai/LiuRLP17, DBLP:conf/iclr/MessaoudMXP0CC24, DBLP:conf/nips/PuGHLHC17, DBLP:journals/tsp/Kassab and its theoretical foundations have been substantially clarified in parallel. Mean-field convergence has been characterized in both the continuous-time [LLN2019, Duncan2019OnTG, NEURIPS2020'16f8e136 and discrete-time [NEURIPS20203202111c, pmlr-v162-salim22a] regimes, finite-particle convergence rates have been established under a variety of assumptions [balasubramanian2024improvedfinitepart, NEURIPS2023'54e5d7af], and the dynamics have been identified with gradient flows on spaces of probability measures [DBLP:conf/nips/Liu17, Duncan2019OnTG], in analogy with the variational formulation of **VariationalFormulationFokker**.

Running through this body of work is a single decisive factor: the kernel. Because it dictates the pairwise interaction between particles, the kernel controls both the trajectory and the limiting behavior of the method (see also Figures 1 and 9). The available convergence results are typically stated as mean-field convergence in KSD, and the extent to which KSD convergence entails weak convergence is itself determined by the kernel [pmlr-v70-gorham17a]. Against this backdrop, the ubiquitous median heuristic [JMLR:v13:gretton12a] is expedient but theoretically unjustified, and its performance is known to deteriorate as the dimension grows [DBLP:conf/iclr/BaEGSSWZ22, DBLP:conf/icml/ZhuoLSZCZ18, DBLP:conf/icml/WangZ018]. Efforts to counteract this degradation in high dimensions have proceeded along two largely orthogonal lines: modifying the dynamics themselves [stein'newton, NEURIPS2019'5dcd0ddd, gong2021sliced, pmlr-v151-liu22a] and learning the update direction with neural networks [langosco2022neural, ZHAO2023118975]. Both introduce additional architectural machinery, leaving open the question of how much can be recovered by kernel selection alone.

Closest to our work is ai'stein'2023, which introduces a mixture-of-kernels framework: a KSD is defined for a weighted linear combination of base kernels, and the mixture weights are learned by maximizing this multiple-kernel KSD. Their construction is nonetheless confined to a finite kernel basis and does not address the continuous parameter optimization we develop here. Our method subsumes their setting as a special case while extending the maximization principle to arbitrary parameterized kernel families, including the dimension-dependent bandwidths that prove essential in high dimensions.

1.2 Contributions.

This work is organized around a single, deliberately isolated research question: can SVGD, stripped of any architectural or algorithmic augmentation, attain competitive performance solely by being equipped with a principled, adaptively chosen kernel? By decoupling kernel selection from the other modifications that populate the SVGD literature, we are able to attribute performance gains unambiguously to the kernel and to quantify how far adaptive kernel learning alone extends the reach of the original method. Our findings answer this question in the affirmative and, in doing so,

recast kernel selection from an incidental hyperparameter choice into the primary lever governing SVGD in high dimensions.

Our contributions are the following.

- (i) **Adaptive kernel selection.** We introduce Ad-SVG, which tunes the kernel online by ascending the KSD between the current particle measure and the target during inference. Unlike the median heuristic, which commits to a single scalar bandwidth, our formulation optimizes an arbitrary vector of continuous kernel parameters—in particular, per-dimension bandwidths—and applies to any parameterized kernel family, at no additional gradient-evaluation cost over vanilla SVGD.
- (ii) **Convergence theory for the adaptive regime.** We analyze Ad-SVG in the discrete-time mean-field setting and lift the fixed-kernel convergence guarantees of prior work to a kernel that is re-selected at every step. We prove that the maximal KSD over the parameterized class converges to zero as the particle distribution approaches the target, both for exact and for approximate KSD maximization. Under a Stein logarithmic Sobolev inequality—which we require of only a single kernel in the class—we further obtain convergence in KL divergence, together with explicit iteration-complexity rates and a bias bound for the case of fixed-accuracy inner maximization.
- (iii) **Empirical validation across dimensions and tasks.** On Gaussian mixtures, an ODE-governed linear inverse problem, high-dimensional Gaussian and Gaussian-process targets, and Bayesian logistic regression, Ad-SVG consistently outperforms the median heuristic and, decisively, alleviates the variance collapse that degrades SVGD as the dimension increases.

2 Mathematical background

We fix throughout a symmetric positive definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ with associated RKHS $\mathcal{H}_0$, and we take $\mathcal{H}$ to be the d -fold Cartesian product of $\mathcal{H}_0$ endowed with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$; the canonical feature map is $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. We write $x \cdot y$ for the Euclidean inner product and $\nabla \cdot$ for the divergence operator. The space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is denoted $\mathcal{P}(\mathbb{R}^d)$, and $\mathcal{P}_p(\mathbb{R}^d)$ its subspace of measures with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, the Wasserstein p -distance is

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ denotes the set of couplings of μ and ν , that is, the probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [IPM] quantify the distance between two measures as the maximal discrepancy of integrals taken over a class of test functions. To render this construction tractable when measuring the deviation of a sample from an intractable target, Stein’s method [bounderrornorma supplies test functions with zero mean under the target: for the Stein operator $\mathcal{S}_{\pi} f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable kernel, one has $\int_{\mathbb{R}^d} \mathcal{S}_{\pi} f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [KernelTestGoodness,

KernelizedSteinDiscrepancy]. Restricting the supremum to the unit ball of $\mathcal{H}$ then yields the *kernelized Stein discrepancy* (KSD),

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. This variational problem admits the closed-form maximizer $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$, where $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$ and $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$, with Φ_k the feature map of the kernel k . Consequently the supremum coincides with the RKHS norm of ψ , giving the equivalent characterization $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$; this identity is what makes the discrepancy computable and, later, differentiable in the kernel parameters.

2.2 Stein Variational Gradient Descent.

Given a target π and a reference measure μ_0 , SVGD transports μ_0 toward π through the iteration $\mu_{n+1} := T_{\sharp} \mu_n$, where $T_{\sharp}$ is the push-forward under the map $T(x) = x + \gamma \psi^{\mu_n}(x)$ and the vector field $\psi^{\mu} = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, $\mu \in \mathcal{P}(\mathbb{R}^d)$, is the direction of steepest descent of the KL divergence. The choice of ψ^{μ_n} is not incidental: the maximizer of Equation (1) shows that $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ is precisely the element of the unit ball of $\mathcal{H}$ that minimizes $\frac{d}{d\gamma} \text{KL}(T_{\sharp} \mu_n \| \pi) \Big|_{\gamma=0}$ [SVGD], with $\text{KL}(\cdot \| \cdot)$ the KL divergence. For this optimal direction the instantaneous rate of KL decrease is exactly the squared discrepancy,

$$\frac{d}{d\gamma} \text{KL}((\text{Id} + \gamma \psi^{\mu_n})_{\sharp} \mu_n \| \pi) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n|\pi). \quad (2)$$

Applying the maps T constructed in this way iteratively to a particle set $\{X_0^i\}_{i=1}^M$ drawn from μ_0 yields the empirical particle updates

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because establishing convergence of SVGD directly in KL divergence is difficult and hinges on restrictive assumptions, recent analyses have instead measured convergence through the kernelized Stein discrepancy [NEURIPS20203202111c, NEURIPS2023'54e5d7af, pmlr-v162-salim22a]. Convergence in KSD is not, however, self-sufficient: it does not by itself imply weak convergence, since the sequence of mean-field measures may fail to be tight—a necessary condition for weak convergence in Polish spaces—and, as the variational problem Equation (1) makes explicit, the RKHS structure, and hence the kernel, dictates how faithfully KSD convergence transfers to weak convergence [pmlr-v70-gorham17a]. The kernel is therefore not merely a tuning knob but the object that determines what convergence in KSD actually certifies, which is precisely why we make it the target of adaptation.

To exploit this dependence rather than suffer it, we formulate our adaptive variant of SVGD over a parameterized family of kernels $\{k_\theta \mid \theta \in \Theta\}$. For each $\theta \in \Theta$ we write KSD_θ for the corresponding kernelized Stein discrepancy, ψ_θ^μ for the induced optimal update direction, and Φ_θ for the associated feature map. Making the kernel dependence explicit in this way lets us select θ

adaptively during inference, so that the kernel’s influence on convergence is actively harnessed and the well-known fragility of a single fixed kernel is removed.

The guiding principle of our approach is to select, at each step, the kernel that maximizes the KSD between the current empirical particle distribution and the target. The rationale is direct and quantitative: under SVGD the instantaneous decrease in KL divergence equals the squared KSD at the current particle measure (Equations (2) and (7)), so maximizing the KSD over the kernel class is exactly maximizing the per-step rate of KL decrease. Geometric analyses of SVGD as a gradient flow corroborate this criterion, showing that kernels inducing larger KSD values yield strictly more favorable convergence behavior when the associated RKHSs are compared [Nskn2021SteinVG, Duncan2019OnTG]. Maximizing the KSD is thus not a heuristic surrogate but the natural optimality criterion implied by the SVGD dynamics themselves.

Whereas vanilla SVGD seeks convergence of the KSD for one fixed kernel, our algorithm targets the largest KSD attainable within the entire kernel class. In contrast to the median heuristic of **SVGD**, which is tied to a specific isotropic kernel, the procedure applies to any parameterized family: at every step it seeks an optimal parameter

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi).$$

Concretely, before each particle update we adjust the kernel parameter by one or more gradient ascent steps on the KSD with step size $s > 0$, and—to control runtime—we optionally refresh the parameter only intermittently rather than at every SVGD step (Algorithm 1), a choice governed by the user-specified flag `paramupdate`. The design is deliberately modular: the base transport step may be replaced by more sophisticated SVGD variants (adaptive step-size schedules, line searches, momentum, or second-order and preconditioned updates) and the gradient ascent on θ by any alternative optimizer, all without altering the kernel-selection mechanism. Our concrete kernel update rests on the closed-form representation [**KernelTestGoodness**, **KernelizedSteinDiscrepancy**]

$$\text{KSD}^2(\mu | \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_{\pi}^k d(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_{\pi}^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \end{aligned} \quad (5)$$

from which the gradient with respect to θ follows by direct differentiation. Two observations establish that this adaptivity comes at essentially no extra cost relative to vanilla SVGD:

- The kernel-parameter ascent reuses the very gradients $\nabla \log \pi(X_n^i)$ already computed for the SVGD step; Ad-SVGd therefore requires no additional evaluations of $\nabla \log \pi$, which are typically the dominant expense.
- The remaining overhead is further reduced by taking only a small number of inner steps `nstepstheta`, or by refreshing the kernel parameter only once per several SVGD steps; when the ensemble size M is large, a subsample of particles suffices to approximate the KSD. In our experiments these controls make the runtime of Ad-SVGd indistinguishable from that of the median heuristic.

Algorithm 1: Ad-SVGD

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps $nsteps, nstepstheta \in \mathbb{N}$

Output: Final particle set $\{X_{nsteps}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```
for  $n = 0$  to  $nsteps - 1$  do
    Decide paramupdate  $\in \{\text{True}, \text{False}\}$ ;
    if paramupdate then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to  $nstepstheta - 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{nstepstheta}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end
```

4 Convergence analysis

We consider a target measure with Lebesgue density $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. To place Ad-SVGD on firm theoretical ground, we extend the recent KSD-based convergence theory of SVGD to the adaptive setting. Specifically, we build on the analysis of [pmlr-v162-salim22a], which operates under Talagrand’s inequality—a condition that holds under mild assumptions on the target and is strictly weaker than the logarithmic Sobolev inequality routinely invoked elsewhere. Adapting that analysis to a kernel that varies across iterations requires only that we impose the assumptions of [pmlr-v162-salim22a] uniformly over all kernel parameters, which we now state.

Assumption 1. *We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.*

Assumption 2. *We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand’s inequality T1, which means that there exists $\lambda > 0$ such that*

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \parallel \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. *We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.*

4.1 Convergence analysis of SVGD under Talagrand's inequality

Throughout we write

$$\mathcal{F}(\mu) := \text{KL}(\mu \parallel \pi).$$

for the KL functional and rely on the following descent inequality. Fix a kernel parameter $\theta \in \Theta$ for which Assumption 3 holds, and define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, for $\gamma, B > 0$, $\alpha > 1$ and $g \in \mathcal{H}_\theta$ satisfying the step-size restriction $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, Proposition 3.1 of [pmlr-v162-salim22a] yields

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta}^2 \quad (6)$$

with $K = (\alpha^2 + L)B$. Specializing this bound to the iterates (3) for a fixed kernel parameter $\theta \in \Theta$ yields the descent condition of [pmlr-v162-salim22a],

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\|) + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1}. \quad (8)$$

The crux of (7) is verifying the step-size condition $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, which Talagrand's inequality (Assumption 2) secures and which in turn licenses (6) with $g = \psi_\theta^{\mu_n}$. Summing (7) then gives $\sum_{n=0}^\infty \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ with $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$, whence

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0.$$

The essential observation is that this argument is entirely local in the kernel: under an adaptive parameter sequence (θ_n) , the descent condition (7) carries over verbatim with θ replaced by the current θ_n ,

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGd.

We first treat the idealized case in which the kernel maximization is solved exactly, so that the adaptive iteration takes the form

$$\begin{aligned} \mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n}) \# \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi), \end{aligned} \quad (9)$$

that is,

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

This exact formulation is available whenever the KSD admits a (unique) closed-form maximizer over Θ . Such is the case for the multiple-kernel SVGD framework of [ai'stein'2023], in which the kernel is a convex combination of base kernels,

$$k_\theta(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1,$$

so our exact analysis covers that method as a particular instance. In the general setting the maximizer θ_n in (9) must be approximated, and we assume throughout that $\arg\max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

For this exact scheme, applying the fixed-kernel descent condition at each step with the maximizing parameter yields the following guarantee, a direct consequence of Theorem 3.2 in [pmlr-v162-salim22a].

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

there exists $c_\gamma > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi) = 0$.*

The more realistic—and more broadly applicable—situation is one in which the KSD maximization is solved only approximately, as is the case for the gradient-ascent inner loop of Algorithm 1. Assuming $\max_{\theta \in \Theta} \text{KSD}_\theta(\mu | \pi) < \infty$ for every $\mu \in \mathcal{P}(\mathbb{R}^d)$, we analyze the alternating scheme

$$\begin{aligned} \theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\text{Id} + \gamma \psi_{\theta_n}^{\mu_n})_\# \mu_n \end{aligned} \tag{10}$$

for a sequence of iterative update rules $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ tasked with maximizing $\text{KSD}_\theta(\mu_n | \pi)$. In Algorithm 1, Ψ_n is the gradient-ascent update on the KSD, but—consistent with the modularity emphasized above—any iterative optimizer with the same asymptotic behavior may be substituted. That behavior is captured by a single, mild requirement on the optimization error.

Assumption 4. *We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and*

$$\max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi) - \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

Under this assumption alone, the maximal KSD over the kernel class is driven to zero.

Theorem 3. *Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [pmlr-v162-salim22a] we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \text{KSD}_{\theta_n}^2(\mu_n|\pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and, in particular, it is a Cauchy sequence. Hence, we have

$$c_\gamma \text{KSD}_{\theta_n}^2(\mu_n|\pi) \leq F(\mu_n) - F(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi) \leq \text{KSD}_{\theta_n}^2(\mu_n|\pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

This guarantee can be sharpened considerably. Under a Stein logarithmic Sobolev inequality, we upgrade the convergence of Ad-SVGD from the KSD to the KL divergence itself; for concrete decay profiles of ε_n we obtain explicit iteration-complexity rates for the mean-field system (10), and the same setting delivers an exact characterization of the bias incurred when the KSD maximization is solved only to a fixed accuracy $\bar{\varepsilon}$ per iteration. The full development is deferred to Appendix B. A notable feature of the adaptive setting is that the log-Sobolev condition need hold for only *one* kernel in the class $\{k_\theta\}$ rather than for a prescribed kernel, which affords substantial flexibility in verifying it in practice.

5 Numerical Experiments

We now evaluate Ad-SVGD empirically, structuring the study to isolate the effect of kernel adaptation across regimes of increasing difficulty. The main text reports two complementary experiments: a one-dimensional Gaussian mixture, which exposes the sensitivity of SVGD to the bandwidth, and a linear inverse problem governed by an ordinary differential equation, which tests the method on a genuinely multivariate posterior. Experiments probing the effect of growing dimensionality on Gaussian and Gaussian-process targets are collected in Appendix A, and Section 5.4 demonstrates the advantage of Ad-SVGD over the median heuristic on the Bayesian logistic regression benchmark established in prior SVGD literature. Throughout, the comparison is deliberately controlled: Ad-SVGD and median-heuristic SVGD share the same transport dynamics, step sizes, and particle counts, so that any difference in approximation quality is attributable to kernel selection alone.

5.1 Kernel parameterization

SVGD is most commonly instantiated with kernels of the form

$$k_h(x, y) = \exp\left(-\frac{\|x - y\|_p^p}{h}\right),$$

where $\|\cdot\|_p$ is the p -norm on $\mathbb{R}^d$ [SVGD, DBLP:conf/iclr/BaEGSSWZ22, Duncan2019OnTG], and the object of selection is the scalar *bandwidth* h . The standard heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, with med the current median pairwise distance between particles, a rule motivated by balancing

the two terms of the SVGD update (3) [SVGD]. Its single scalar collapses all coordinate directions onto one length scale—the very rigidity that our adaptive scheme is designed to remove.

To realize the flexibility of the adaptive method, we adopt anisotropic product kernels

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

parameterized by a bandwidth vector $h = (h_1, \dots, h_d)$, thereby assigning an independent length scale to each coordinate. The derivatives required to instantiate Algorithm 1 for this family (via Equations (4) and (5)) are available in closed form. For a controlled comparison we also evaluated a per-dimension variant of the median heuristic that assigns each coordinate its own naive median; this variant failed to match the adaptive scheme and exhibited pronounced variance collapse, confirming that the gains stem from KSD-driven selection rather than from anisotropy alone.

The same construction points to more scalable kernel families for higher dimensions. Spectral or low-rank parameterizations can adapt to the dominant directions of variability without the cost of a full product kernel, and Matérn or mixed-product constructions afford additional control over smoothness and anisotropy; each fits directly into the parameterized-kernel formulation of Section 3 and the accompanying convergence theory without modification.

5.2 Toy example

We begin with the one-dimensional benchmark of SVGD, the two-component Gaussian mixture $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. Initializing the particles from $\mathcal{N}(0, 1)$ with $M \in \{50, 200, 500\}$, we run SVGD for 10^4 steps at step size 1, using the kernel above with $p = 1$ across a range of fixed bandwidths h . Sample quality is measured by the Wasserstein 1-distance $\mathcal{W}_1$, computed from the explicit formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see **StatisticalAspectsWasserstein**) against an exact reference sample of size 10^5 . Figure 1 reports the outcome and makes the central difficulty concrete: SVGD attains a small $\mathcal{W}_1$ only for bandwidths within a narrow interval, with accuracy deteriorating steeply on either side. Even in one dimension, then, performance is governed almost entirely by h , so that a principled selection strategy is not merely helpful but decisive—the gap that Ad-SVGD is designed to close.

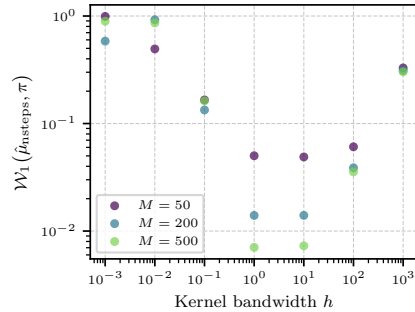


Figure 1: Final Wasserstein 1-distance $\mathcal{W}_1$ to the target for the one-dimensional Gaussian mixture, obtained by SVGD with a range of fixed bandwidths h (lower is better). Performance is competitive only within a narrow band, and degrades sharply outside it, exposing the bandwidth sensitivity that motivates adaptive selection.

5.3 Linear inverse problem based on ODE

We now turn to a multivariate posterior arising from a partial-observation inverse problem, adapted from [Weissmann2022Optimization]. Consider the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

together with the associated inverse problem of recovering the right-hand side $u(\cdot) \in L^2([0, 1])$ from discrete, noisy observations of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. The observation model is

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ is observational noise and the forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is defined by $\mathcal{O} \circ H^{-1}$, with $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ being the observation operator at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation of the inverse problem, we place a Gaussian process (GP) prior on u through the truncated Karhunen–Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and the coefficients are independent, $x_k \sim \mathcal{N}(0, \lambda_k)$ with $\lambda_k = 50k^{-2}$. Inference thus reduces to estimating the KL coefficient vector $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ under the prior $\mathcal{N}(0, \Gamma_0)$, $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. With Gaussian observation noise $\varepsilon \sim \mathcal{N}(0, \Gamma)$, Bayes' rule gives the posterior density

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

In the implementation we replace H by a numerical discretization of Equation (11) on a grid of mesh size 2^{-8} and consider the fully observed system ($N_{\text{obs}} = 2^8$). We retain $N_x = 16$ KL terms for u and set the noise covariance to $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. Reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

We compare SVGD with the median heuristic—henceforth Med-SVG—against Ad-SVG across ensemble sizes $M \in \{50, 100, 200\}$, using the kernels of Section 5.1 with $p = 1$. Both methods are run for $4 \cdot 10^5$ iterations at particle step size 10^{-3} with the AdaGrad-based step-size control recommended by **SVG**; Ad-SVG updates the bandwidth vector with step size 10^{-5} once per 100 particle updates. Under this schedule the two methods are computationally indistinguishable, so the comparison isolates the effect of kernel adaptation at matched cost.

Figure 2 shows the GP reconstruction for a representative seed: both methods recover the posterior mean accurately, but only Ad-SVG reproduces the posterior uncertainty, whereas Med-SVG collapses it. To quantify approximation quality we use the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the Gaussian $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ fitted to the particle sample mean $\hat{\mu}$ and covariance $\hat{\Sigma}$ [StatisticalAspectsWasserstein]. Because π is multivariate normal, this admits the explicit form

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

with μ_π, Σ_π the posterior mean and covariance; we additionally compare the marginal variances of the final particles against the posterior. Figure 3 aggregates these results over 56 random seeds

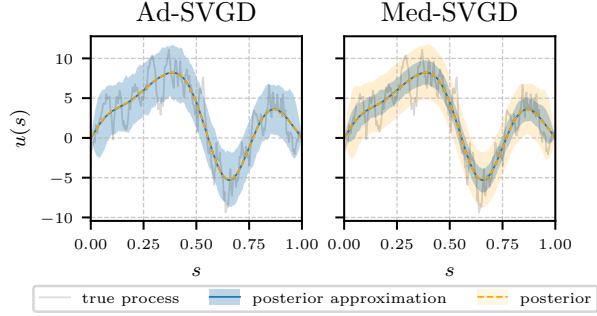
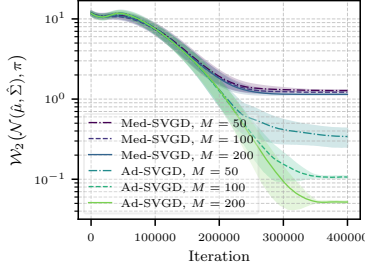
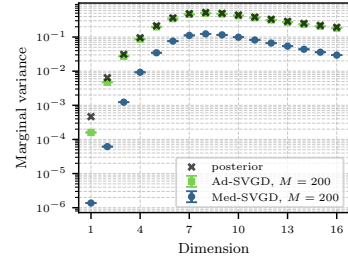


Figure 2: Gaussian-process reconstruction for the ODE-based inverse problem (mean and 90% confidence interval). Both methods recover the posterior mean, but only Ad-SVG reproduces the posterior uncertainty band; median-heuristic SVG collapses it.

(the posterior covariance is independent of $\bar{y}$). Ad-SVGD attains uniformly smaller $\mathcal{W}_2$ error than Med-SVGD, which systematically underestimates the posterior uncertainty; moreover, and in sharp contrast to Med-SVGD, the accuracy of Ad-SVGD continues to improve as the ensemble grows beyond 50 particles—evidence that adaptation lets the method exploit additional particles rather than saturate. Figure 4 traces the bandwidths selected by Ad-SVGD: the component-wise bandwidths stabilize appreciably faster than the approximation error, and the converged values differ clearly across coordinates, being negatively correlated with the corresponding marginal variances. The method thus discovers an interpretable, anisotropic length scale that the single median bandwidth cannot represent.

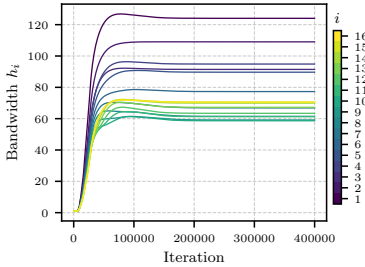


(a) Convergence of the $\mathcal{W}_2$ approximation error

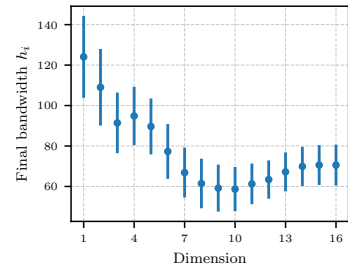


(b) Marginal variances vs. the posterior

Figure 3: ODE-based inverse problem: Med-SVGD versus Ad-SVGD (mean and 95% confidence interval over 56 random seeds). Ad-SVGD attains a smaller $\mathcal{W}_2$ error and recovers the marginal variances, whereas Med-SVGD underestimates them; the Ad-SVGD error also keeps improving with the ensemble size.



(a) Per-dimension bandwidth evolution (mean)



(b) Final bandwidths (mean and 95% conf. interval)

Figure 4: Bandwidths selected by Ad-SVGD for the ODE-based inverse problem, aggregated over 56 random seeds. The component-wise bandwidths stabilize faster than the approximation error and settle at clearly distinct, anisotropic values negatively correlated with the marginal variances.

5.4 Bayesian logistic regression

Following **SVGD** and **pmlr-v151-liu22a**, we evaluate the methods on a Bayesian logistic regression (BLR) model for the Covertypes data set [**covertypes**], a standard SVGD benchmark. The regression weights w receive a hierarchical Gaussian prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim$

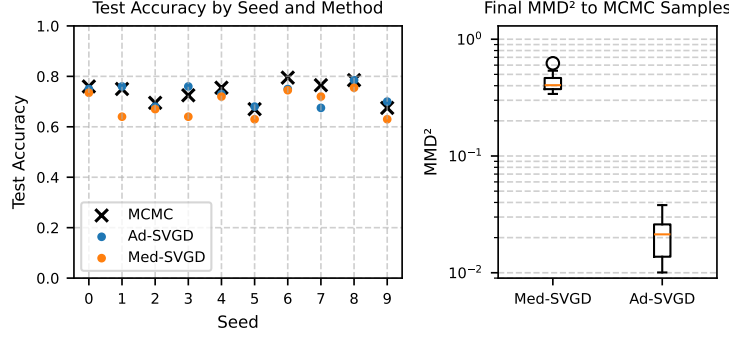


Figure 5: Bayesian logistic regression: approximation quality of the SVGD particles, measured by test prediction accuracy (left) and squared maximum mean discrepancy MMD^2 to the NUTS reference (right), over 10 random draws. Ad-SVGd attains higher accuracy and a smaller MMD^2 than Med-SVGd across nearly all seeds.

Gamma(1, 0.01), and the target of inference is the joint posterior of $x = [w, \alpha]$. As in **pmlr-v151-liu22a**, we obtain a high-quality reference posterior from the No-U-Turn Sampler (NUTS) of **nuts**, against which the approximation quality of Med-SVGd and Ad-SVGd is assessed. Following **pmlr-v151-liu22a**, we test on random subsets of size 1000 and aggregate over 10 draws, reporting the classification accuracy on a held-out test set obtained from the particle mean. As Figure 5 shows, Ad-SVGd dominates Med-SVGd across nearly all seeds, simultaneously achieving higher test accuracy and closer agreement with the MCMC reference—so that better posterior fidelity translates directly into better predictive performance.

As a direct measure of posterior fidelity, the same figure reports the squared maximum mean discrepancy (MMD^2) [**MMDref**] to the MCMC samples, on which the two methods differ markedly. Figure 7 corroborates this: the final-particle covariance matrices, averaged over the 10 seeds (following the visualization of **pmlr-v151-liu22a**), reveal that Med-SVGd severely underestimates the posterior uncertainty, while Ad-SVGd reproduces it far more faithfully. Figure 6 explains the mechanism: starting from a uniform, uninformed initialization (1 in every component), Ad-SVGd adaptively drives the component bandwidths across several orders of magnitude to values suited to the posterior geometry—an adjustment entirely beyond the reach of a single scalar bandwidth.

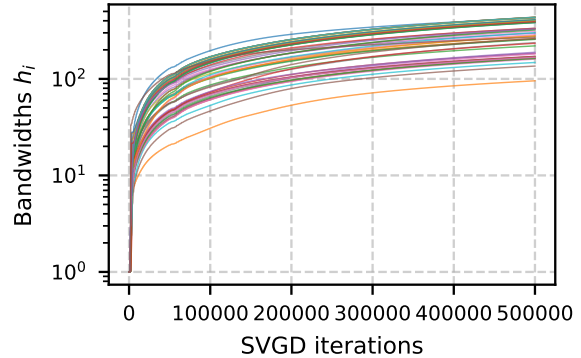


Figure 6: Bandwidth evolution for the BLR example. From an uninformed initialization (1 in every component), Ad-SVGd recovers component-wise bandwidths that differ from the initialization—and from one another—by several orders of magnitude.

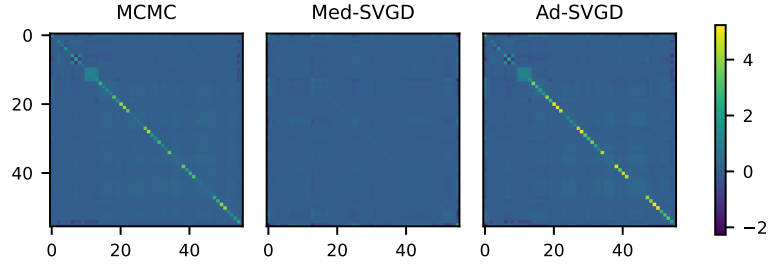


Figure 7: Covariance matrices of the final SVGD particle sets compared to the MCMC reference (mean over 10 seeds). Med-SVGd collapses the off-diagonal structure and underestimates the variances, whereas Ad-SVGd closely matches the reference covariance.

6 Limitations and outlook

The principal limitation of our theory is its reliance on Assumption 4, which posits that the inner optimizer eventually closes the gap to the maximal KSD. Establishing when the alternating gradient-ascent scheme of Algorithm 1 provably satisfies this condition is the subject of ongoing work; while we do not yet certify it in general, the strong and consistent empirical performance across all our experiments indicates that it is met in the regimes of practical interest. We stress that this assumption concerns only the inner kernel optimization, not the transport dynamics, whose convergence we establish unconditionally given it.

Because our analysis targets the original SVGD dynamics, the adaptive kernel-selection mechanism is deliberately orthogonal to modifications of the transport step and composes with them directly. Combining it with recent variants such as sliced SVGD [gong2021sliced], Grassmann SVGD [pmlr-v151-liu22a], or Stein transport [nsken2024steintransportbayesianinference] is therefore a natural and promising avenue, offering the prospect of compounding the gains from adaptive kernels with those from improved dynamics.

A More numerical experiments

A.1 Gaussian mixture models

We first return to the mixture of Section 5.2. To make the bandwidth sensitivity of Figure 1 tangible at the level of the distribution, Figure 9 overlays the final particle histograms on the target density for $h \in \{0.001, 1, 1000\}$, illustrating how markedly the fixed-bandwidth approximation deteriorates away from the well-tuned regime. We then compare Med-SVGD and Ad-SVGD directly, matching the final particle distributions after 10^4 iterations at step size 1 across a range of ensemble sizes M .

Figure 8 reports the $\mathcal{W}_1$ -distance between the final empirical distribution and the target for $M = 10, 20, 50, 100, 200, 500$. As expected, accuracy improves monotonically with M for both methods, and both reach $\mathcal{W}_1 < 0.01$ at $M = 500$. In this benign one-dimensional setting adaptive selection is thus on par with a well-tuned median heuristic—establishing that the substantial gains reported elsewhere emerge specifically where the fixed heuristic breaks down, namely in higher dimensions, rather than reflecting an across-the-board advantage that would be easy to overstate.

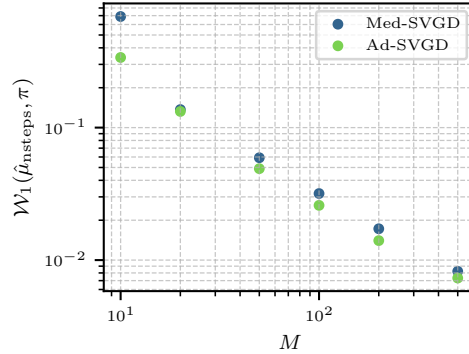


Figure 8: Approximation error ($\mathcal{W}_1$) for the 1D Gaussian mixture as a function of the particle count. Med-SVGD and Ad-SVGD perform comparably, both reaching $\mathcal{W}_1 < 0.01$ at $M = 500$.

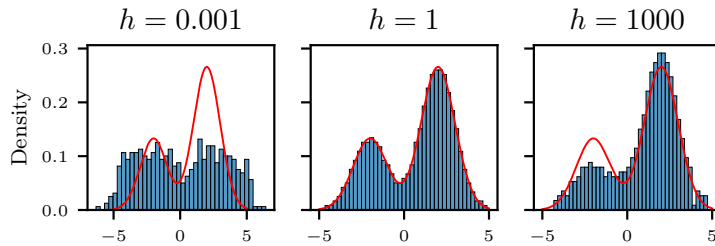


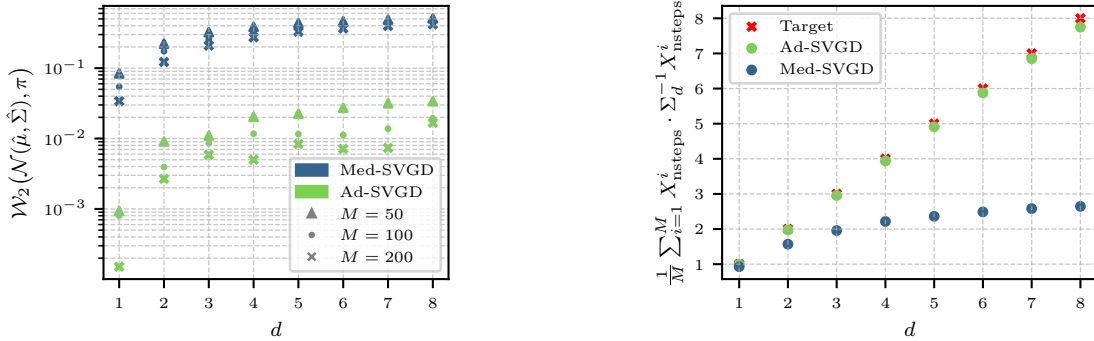
Figure 9: Histograms of the final particle set for fixed bandwidths $h = 0.001, 1, 1000$, with the target density π in red. Only the intermediate bandwidth recovers the bimodal target; the extreme values either collapse the particles or oversmooth them, visualizing the sensitivity quantified in Figure 1.

In one dimension, both methods approximate the target well: adaptive bandwidth selection performs on par with the median heuristic without conferring a decisive advantage in this regime. This parity is informative rather than negative—it delineates the boundary of the median heuristic’s competence and sets up the higher-dimensional experiments below, where that competence fails and adaptation becomes essential.

A.2 Scaling dimension: multivariate normal distribution

To probe the high-dimensional regime systematically, we consider the anisotropic Gaussian targets $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$, whose sharply differing coordinate scales are precisely what a single scalar bandwidth cannot accommodate. The particles are initialized from $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$ and both algorithms are run for 10^4 iterations at step size 0.1, reducing the step size and lengthening the run whenever numerical stability requires.

We track the approximation quality of the final particle set for Med-SVGD and Ad-SVGD as d grows. Since the $\mathcal{W}_2$ -distance between Gaussians is available in closed form (see [StatisticalAspectsWasserstein]) we adopt as our quality measure the $\mathcal{W}_2$ -distance between π_d and the Gaussian $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ fitted to the empirical particle mean $\hat{\mu}$ and covariance $\hat{\Sigma}$ [StatisticalAspectsWasserstein]; following the convention of the Python Optimal Transport library [PythonOptimalTransport] used in our computation, we refer to this as the *Bures–Wasserstein distance*. Figure 10a plots this measure for $M = 50, 100, 200$ as a function of dimension, and the separation is unambiguous: Ad-SVGD significantly outperforms Med-SVGD at every value of d .



(a) Bures–Wasserstein error for $M \in \{50, 100, 200\}$. (b) χ^2 test statistic for $M = 200$ (target value d).

Figure 10: Multivariate normal targets of increasing dimension. Ad-SVGD attains a smaller Bures–Wasserstein error at every d (a), and its χ^2 statistic tracks the ideal value d while that of Med-SVGD diverges as d grows (b), evidencing progressive variance underestimation by the median heuristic.

The Bures–Wasserstein distance is relatively insensitive to the small variances in the trailing components of π_d , so to expose how well each method captures uncertainty in exactly those directions we turn to the test statistic $X \cdot \Sigma_d^{-1} X$, which for $X \sim \mathcal{N}(0, \Sigma_d)$ is χ^2 -distributed with d degrees of freedom and hence has expected value d (cf. [Metropolisadjustedinteracting]). Evaluating its mean over the $M = 200$ particles and comparing to the ideal value d in Figure 10b yields a clean diagnostic: the Med-SVGD statistic drifts progressively further below d as the dimension grows, whereas the Ad-SVGD statistic remains close to d throughout. Med-SVGD therefore fails to represent the higher-dimensional targets, while Ad-SVGD handles them accurately.

Tables 1 and 2 make this quantitative at the level of individual coordinates, listing for each d the marginal variances of the particle set alongside the target variances (first row) for Med-SVGD and Ad-SVGD respectively. Ad-SVGD recovers the target marginal variances accurately across all components and all dimensions, whereas Med-SVGD substantially underestimates them—in the leading component the Med-SVGD variance falls from 0.93 at $d = 1$ to 0.48 at $d = 8$, while Ad-SVGD stays near 0.97–1.00—consistent with the χ^2 diagnostic of Figure 10b.

Table 1: Marginal variances of the final Med-SVGD particle distribution ($M = 200$) for the d -dimensional Gaussian targets; the target marginal variances π_d appear in the first row. Med-SVGD underestimates the variances across all components, with the shortfall growing as d increases.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Marginal variances of the final Ad-SVGD particle distribution ($M = 200$) for the d -dimensional Gaussian targets; the target marginal variances π_d appear in the first row. Ad-SVGD recovers the target variances closely in every component and dimension.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

Finally, we examine the marginal particle distributions for $M = 200$ and $d = 8$ in detail. Normalizing each coordinate by the inverse of the corresponding target marginal standard deviation (equivalently, scaling the i -th component by i) renders every marginal of π_d a standard normal. The resulting histograms are shown in Figures 11 and 12 for Med-SVGD and Ad-SVGD, each against the standard Gaussian density. Ad-SVGD reproduces all eight marginals faithfully, whereas Med-SVGD visibly understates their spread; the Q-Q plots of Figures 13 and 14, comparing the normalized marginal quantiles against the standard-normal quantiles, tell the same story component by component.

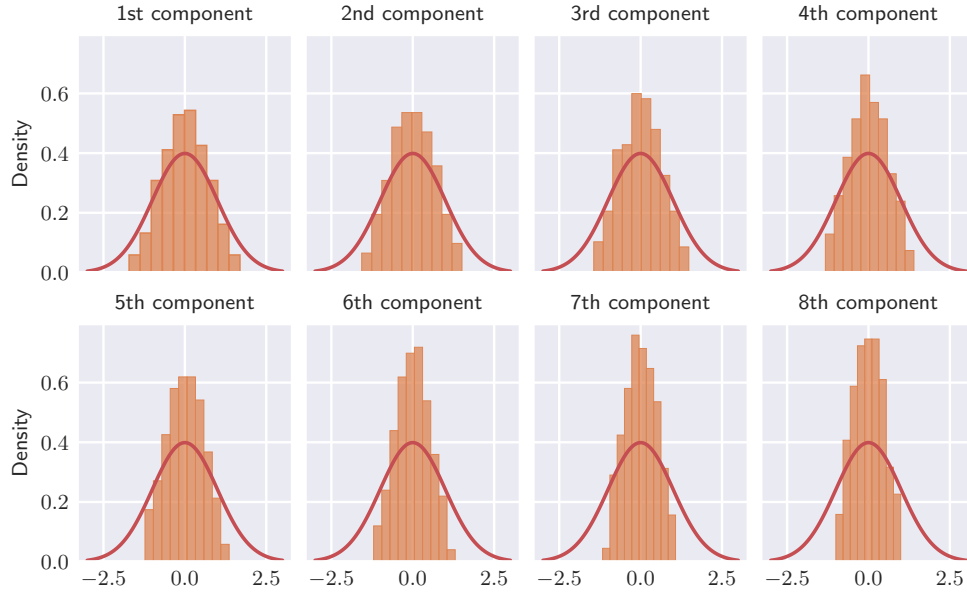


Figure 11: Component-wise histograms of the final Med-SVG particles for the eight-dimensional target ($M = 200$), rescaled by the marginal target standard deviations so that each target marginal is standard normal (shown for comparison). The particle marginals are visibly too concentrated, confirming systematic variance underestimation.

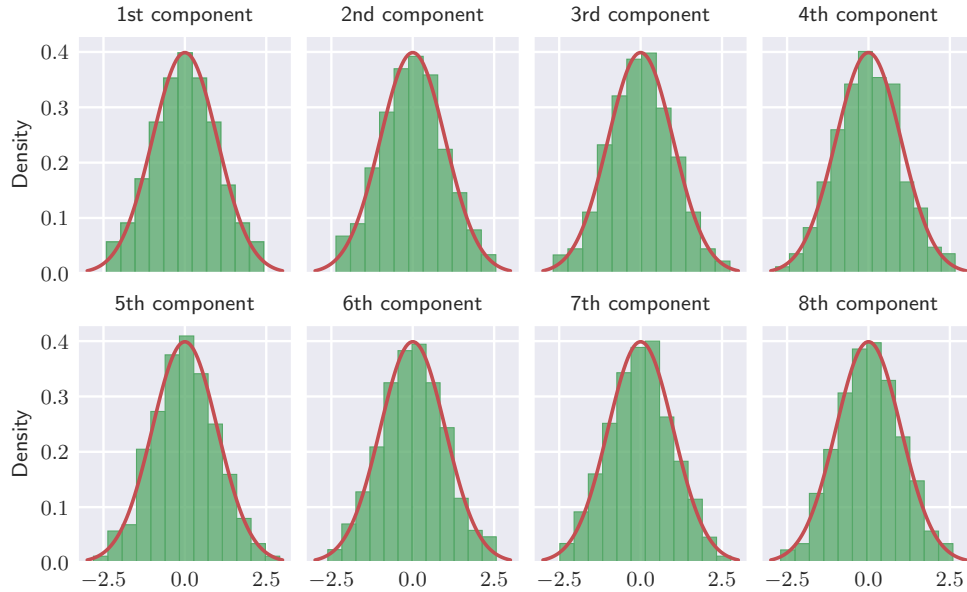


Figure 12: Component-wise histograms of the final Ad-SVG particles for the eight-dimensional target ($M = 200$), rescaled by the marginal target standard deviations so that each target marginal is standard normal (shown for comparison). All eight marginals are captured accurately.

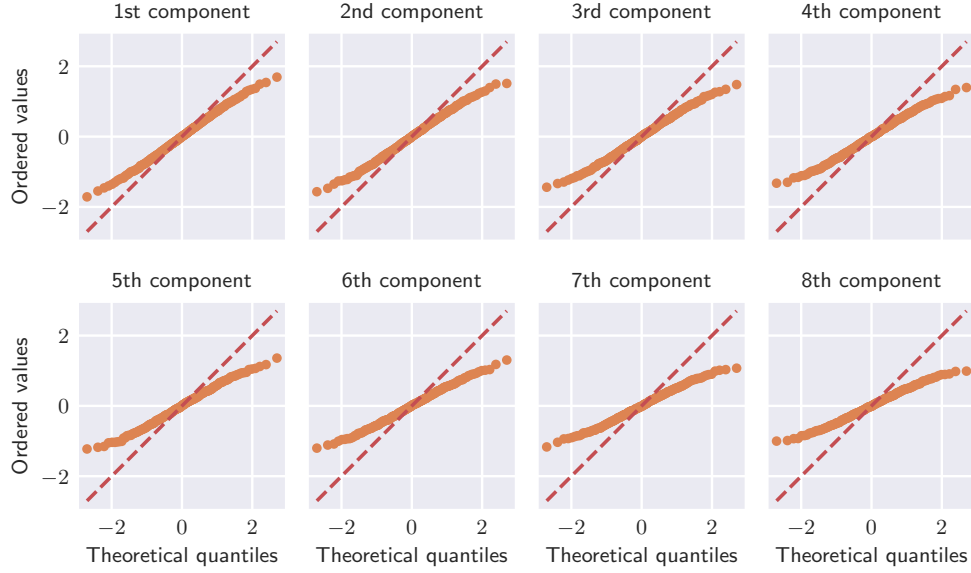


Figure 13: Q–Q plots of the normalized marginals of the final Med-SVGD particles for the eight-dimensional target ($M = 200$) against the standard normal, with the identity line of unit slope through the origin shown for reference. Departures from the identity line reflect the underestimated marginal spread.

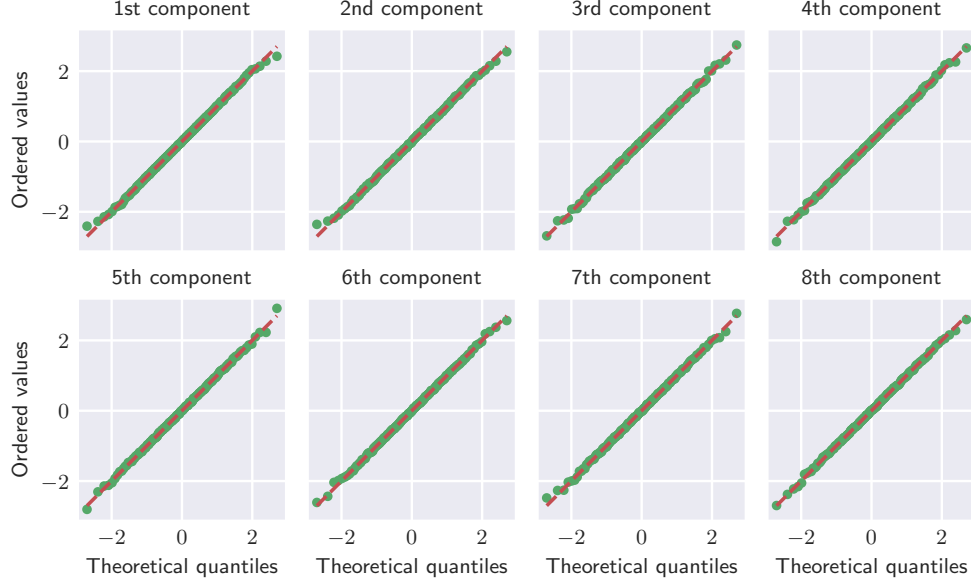


Figure 14: Q–Q plots of the normalized marginals of the final Ad-SVGD particles for the eight-dimensional target ($M = 200$) against the standard normal, with the identity line of unit slope through the origin shown for reference. The marginal quantiles align closely with the identity line across all components.

A.3 Gaussian process inference

As a further high-dimensional test, we study the Gaussian-process (GP) inference task of [reich21], which lets us vary both the latent dimension and the number of observations independently. We consider a GP on $[0, 1]$ represented by a truncated Karhunen–Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ collects independent Gaussian coefficients $x_k \sim \mathcal{N}(0, k^{-2})$. Observing the process at N_y equispaced points in $[0, 1]$ and inferring the coefficients x_k yields, for fixed N_x and N_y , a linear-Gaussian inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$, $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with Σ diagonal and entries k^{-2} , $k = 1, \dots, N_x$, and the noise is independent Gaussian, $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. Reference observations $\bar{y} \in \mathbb{R}^{N_y}$ are constructed by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

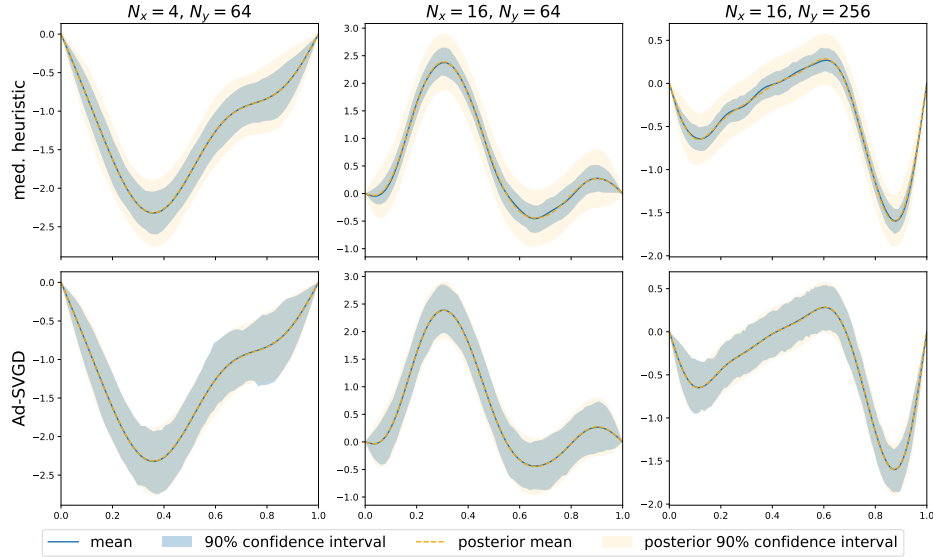


Figure 15: Estimated processes from median-heuristic SVGD and Ad-SVGD ($M = 100$ particles) for three model configurations, compared to the posterior. As the latent dimension grows, the median heuristic collapses the posterior uncertainty band, whereas Ad-SVGD tracks it; the effect is consistent across the number of observations.

We sample from the resulting posterior $X | Y = \bar{y}$ with SVGD and again pit the median heuristic against the adaptive approach. Median-heuristic SVGD uses the isotropic kernel $k(x, y) = \exp(-\|x - y\|_1/h)$, while Ad-SVGD uses the anisotropic product kernel $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with a distinct bandwidth h_i per coordinate; for the $N_x = 16$ configurations we employ the AdaGrad-based step-size control of [SVGD].

Figure 15 contrasts the estimated processes with the posterior for the configurations $(N_x, N_y) = (4, 64)$, $(16, 64)$, and $(16, 256)$. As the latent dimension increases, median-heuristic SVGD progressively underestimates the posterior variance, whereas Ad-SVGD sustains an accurate approximation, and this pattern persists across the different observation counts—indicating that the advantage is driven by the latent dimension rather than by the amount of data. Table 3 quantifies the comparison through the trace of the particle covariance against the true posterior: the median heuristic severely understates the total variance, while Ad-SVGD recovers it closely in every configuration.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the final particle covariance ($M = 100$ particles) against the theoretical posterior trace across latent dimensions N_x and observation counts N_y , averaged over 25 runs. Ad-SVGD closely matches the theoretical trace in every configuration, whereas the median heuristic recovers less than half of it.

B Refined theoretical analysis

We now strengthen the convergence result under a Stein log-Sobolev inequality, which couples the KL divergence to the KSD. This inequality was introduced by **Duncan2019OnTG**, who also discussed conditions under which it holds. A distinctive benefit of the adaptive setting is that we may substantially weaken this requirement: rather than demanding it of a prescribed kernel, we ask only that *some* kernel in the class satisfy it, as the maximization over θ automatically selects the most favorable member. This is made precise in the following assumption.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \mid \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

This property lets us quantify convergence directly in KL divergence. Notably, it delivers an explicit error bound on $\mathcal{F}(\mu_n)$ even under the merely approximate condition $\varepsilon_n \leq \bar{\varepsilon}$, that is, without requiring the inner optimization error to vanish—so that a fixed-accuracy kernel maximization already suffices to control the bias.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) - \text{KSD}_{\theta_n}^2(\mu_n \mid \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

Proof. Similar to the proof of Theorem 3, under Assumption 5 and the conditions on α, γ , we have

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$, we deduce the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We consider the three different cases separately:

1. Under $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows by the computation of the geometric series

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1 - \rho}.$$

2. Assuming that $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$ we have

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1 - \rho/q} = q^n \frac{1}{q - \rho}$$

where we have used the formula for geometric series. Similarly, in the case of $q < \rho$, we can directly bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1 - q/\rho} = \rho^n \frac{1}{(\rho - q)}.$$

Finally, in the case $q = \rho$, we simply bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ and deduce the claim line by line as before.

3. In the setting of $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here, we have used $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□